

TANMAY HINGE

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Minneapolis, Minnesota, USA · Mumbai, Maharashtra, India

Application for a Doctoral Position · Research Group on AI Safety and Societal Impacts (Dr Paul Röttger) · Hasso Plattner Institute

RESEARCH INTERESTS

Safety and evaluation of large language model systems: measuring whether a model's output is actually grounded in the evidence it cites, and building evaluation harnesses that make ungrounded claims visible. Agentic and retrieval-grounded LLM pipelines with human-in-the-loop assessment. Representation probing and the question of whether a reported end-task metric reflects what a system has actually learned. Calibrated uncertainty and honest abstention. Adversarial and threat-model reasoning carried over from formally verified cryptography. Reproducible, seeded experimental pipelines at GPU scale.

EDUCATION

M.S. Computer Science, California State University, Chico, USA 2022 – May 2024

- Capstone: machine-learning analysis of hiring outcomes; supervised classifier comparison over survey-elicited features, with dimensionality reduction, tuning, and an explicit interpretability-versus-accuracy analysis.
- Coursework: Applied Machine Learning, Applied Computer Vision, Applied Graph Theory, Research Methods in Computer Science, Algorithms and Computability, Advanced Computer Security.

B.E. Computer Engineering, Pillai College of Engineering, University of Mumbai, India 2018 – Jun 2022

- Coursework: Natural Language Processing, Machine Learning, Artificial Intelligence and Soft Computing, Digital Signal and Image Processing, Distributed Computing, Theory of Computer Science.

RESEARCH AND PROJECTS

Does Masked Pretraining Learn the Structure of Mixed Spectra? · [code and report](#) 2026

- Built a controlled environment with exact ground truth (frozen twelve-compound library, Lorentzian lineshapes, J-coupling, 2048-point grid), pretrained a 544k-parameter patch transformer with span masking on 20,000 unlabelled signals, and probed the frozen encoder.
- Result: end-task accuracy after full fine-tuning converges across initialisations, while linear probes on the frozen encoder show a large gap. Two standard measurements of the same system disagree, and which one is reported determines the conclusion drawn about what pretraining learned.
- Isolated the cause through a difficulty sweep and a pretext-target ablation: an oracle target recovers only 2% of the gap, locating the failure in the pretext objective rather than in optimisation. Config-driven, seeded, 46 tests.

Calibrated Uncertainty and Generalisation Regimes in Learned Molecular Couplings · [code and report](#) 2026

- Showed that a single reported error figure hides two entirely different regimes: on unseen pairs of familiar units the model reaches 0.008 MAE, while on genuinely novel units it is roughly 600 times worse.
- Made the distinction machine-detectable with a Gaussian process whose posterior variance is small exactly where the model is reliable and large where it is not, so the system reports its own limits instead of extrapolating silently. Exploited Kronecker structure to cut an $O(n^6)$ solve to two 200×200 eigendecompositions.

Formally Verified, Constant-Time Cryptography 2026

- Formalised ElGamal IND-CPA security in EasyCrypt as a reduction to the Decisional Diffie-Hellman assumption, with exploratory work on the Fujisaki-Okamoto transform in the classical random-oracle model.
- Implemented ML-KEM-768 in memory-safe Rust, verified memory safety with Kani (bounded model checking) and constant-time execution of the compiled assembly with dduct.

Reinforcement Learning for Contact-Rich Control 2024

- Implemented and compared PPO and SAC with custom reward shaping in MuJoCo and Isaac Gym; SAC reached 78% task success versus 61% for PPO at roughly 40% fewer samples, and held 65% under domain randomisation. Python, PyTorch.

PROFESSIONAL EXPERIENCE

Founder & ML Engineer, Regace AI Feb 2026 – present

- Built a retrieval-augmented generation engine over regulatory corpora end to end: structure-aware ingestion, hybrid retrieval with reranking, and citation-grounded generation in which every claim traces to a source passage.
- Built the hallucination evaluation harness that gates it, scoring generated claims against retrieved evidence so unsupported statements are caught before they reach a user. Running with pharmaceutical pilot clients.

Co-Founder & Technical Lead, Merit Labs May 2026 – present

- Built an LLM analytics platform end to end (Next.js, PostgreSQL, containerised CI) with a human-in-the-loop grading pipeline, turning raw student responses into individualised reports and iterating the pipeline against expert human judgement.

- Deployed through school partnerships, where the measured question is how an LLM system behaves in the hands of real users rather than on a static benchmark.

Computer Science Faculty, Inver Hills Community College, Minnesota, USA

Aug 2024 – Jun 2026

- Full-time faculty: designed and taught four core CS courses (Introductory CS, C++, Python, Java); mentored 100+ students and contributed to an Applied AI certificate curriculum.

Software Engineer Intern, VettX Inc., California, USA

May 2023 – Aug 2023

- Built a client-onboarding platform (React, Vidyard API) and an AI-assisted Chrome extension that accelerated seller outreach by roughly 40%.

REPORTS & OPEN SOURCE

- Hinge, T. (2026). Does masked pretraining learn the structure of mixed spectra? Technical report and code. github.com/tanmayhinge/spectral-disentangling
- Hinge, T. (2026). Predicting excitonic couplings from Coulomb-matrix kernels. Technical write-up and code. github.com/tanmayhinge/bimolecular-ml
- HoneyScript: a minimal interpreted programming language with its own context-free grammar. Shipped iOS applications (Swift) totalling roughly 5,000 users worldwide.

TECHNICAL SKILLS

- **LLM Systems & Evaluation:** retrieval-augmented generation (structure-aware ingestion, hybrid retrieval, reranking, citation grounding), agentic pipelines, hallucination and groundedness evaluation harnesses, human-in-the-loop assessment loops; local and hosted serving with vLLM, llama.cpp, and Ollama; fine-tuning with Unsloth, Axolotl, and PEFT.
- **ML & Deep Learning:** PyTorch, TensorFlow, transformer and sequence models, self-supervised and masked pretraining, representation probing, kernel methods and Gaussian processes, calibrated uncertainty, reinforcement learning (PPO, SAC), scikit-learn, MLflow.
- **AI & Security:** formal verification with EasyCrypt and Kani, constant-time analysis with ducedt, threat modelling and adversarial reasoning, memory-safe systems programming in Rust.
- **Scientific Computing & Systems:** NumPy, SciPy, Matplotlib, pytest; GPU-accelerated training; reproducible, seeded, config-driven pipelines; Python (production-grade), C++, Rust, Java, TypeScript; Docker, Kubernetes, CI/CD, Git; PostgreSQL, MongoDB; AWS, GCP, Azure; LaTeX.
- **Languages:** English (fluent), Hindi and Marathi (native), German (in study), Swedish (beginner).

ACADEMIC REFEREES

Samuel B. Siewert, Professor of Computer Science, California State University, Chico.

sbsiewert@csuchico.edu · Graduate Course Instructor, Applied Computer Vision.

Richard C. Tillquist, Assistant Professor of Computer Science, California State University, Chico.

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David Zeichick, Instructor, Department of Computer Science, California State University, Chico.

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